

Goldman Sachs Consumer Finance Data Analysis

SQL Capstone Project — Risk Assessment & Portfolio Health

An end-to-end audit of a large-scale consumer loan dataset to uncover regional growth trends, default risks, and financial performance using Advanced SQL.

Scope	Methods	Focus
Large-scale consumer loan dataset analysis	Advanced SQL techniques & data auditing	Regional trends, default risk & portfolio health

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Regional Customer Analysis

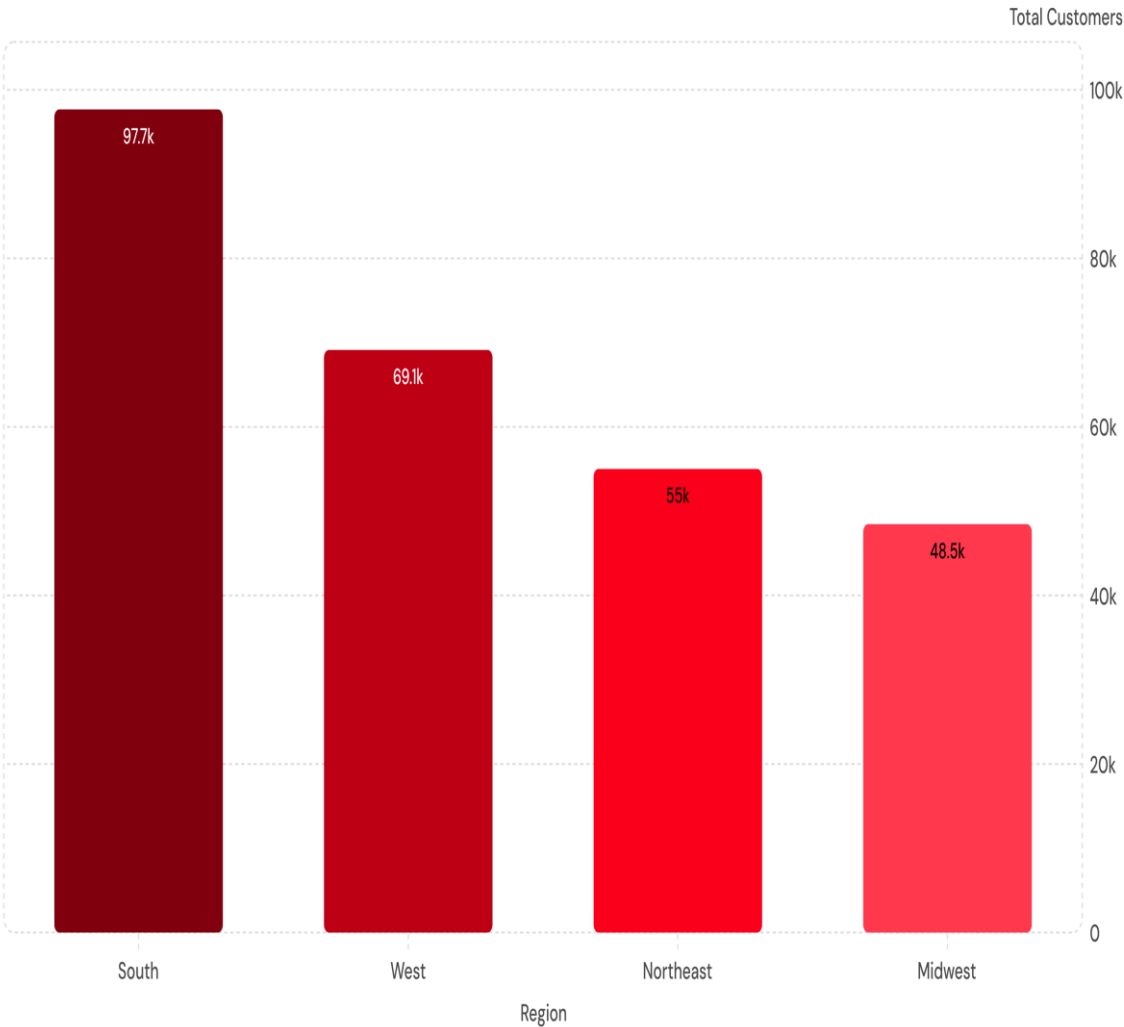
QUESTION 1

SQL Query

```
SELECT
    sr.region,
    COUNT(DISTINCT c.customer_id) AS total_customers,
    ROUND(AVG(c.avg_cur_bal), 2) AS avg_financial_balance
FROM customer c
LEFT JOIN state_region sr
    ON c.addr_state = sr.state
WHERE sr.region IS NOT NULL
GROUP BY sr.region
ORDER BY total_customers DESC;
```

SQL Output

	region	total_customers	avg_financial_balance
►	South	97683	13826.57
	West	69137	15006.51
	Northeast	55014	12731.77
	Midwest	48465	12506.14



Key Takeaway: The South region leads with 97,683 customers — reflecting stronger lending demand and deeper market penetration. The West commands the highest average balance at \$15,006.51, suggesting wealthier borrower profiles despite lower headcount.

Loan Approval Efficiency

QUESTION 2

SQL Query

```
SELECT
    COUNT(loan_id) AS total_funded_loans,
    SUM(CASE WHEN loan_status IN ('Fully Paid', 'Current') THEN 1
    ELSE 0 END) AS successful_loans,
    ROUND(
        SUM(CASE WHEN loan_status IN ('Fully Paid', 'Current') THEN
        1 ELSE 0 END) * 100.0 / COUNT(loan_id),
        2
    ) AS success_rate_percentage
FROM loan;
```

Query Output

270K

Total Funded Loans

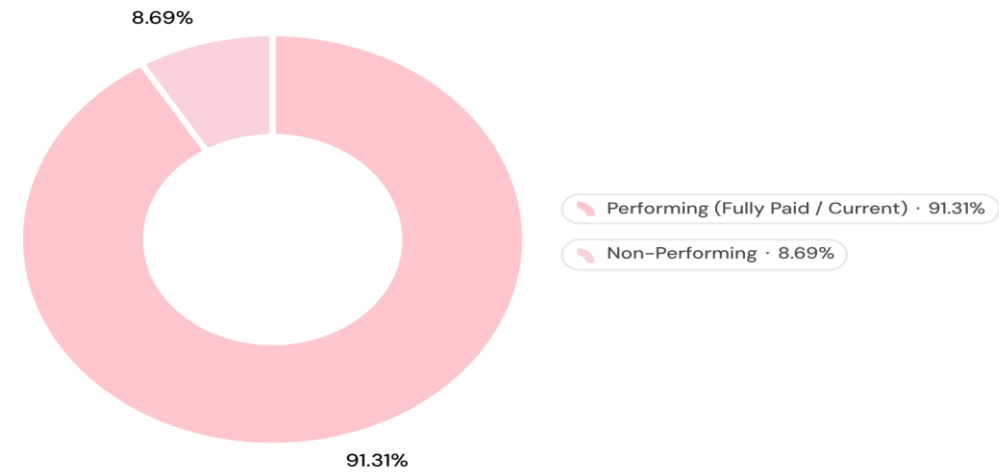
246K

Successful Loans

91.31%

Success Rate

Portfolio Performance — Performing vs. Non-Performing



	total_funded_loans	successful_loans	success_rate_percentage
	270299	246822	91.31

🔑 **Key Takeaway:** With a **91.31% success rate** across 270K+ loans, the portfolio demonstrates strong repayment discipline and limited default exposure — a foundational metric for credit health benchmarking.

High Default Risk Identification

QUESTION 3

```
SQL Query

SELECT
  l.loan_id,
  c.customer_id,
  l.grade,
  c.annual_inc,
  l.loan_amount,
  c.avg_cur_bal AS proxy_credit_health,
  CASE
    WHEN l.loan_amount > (c.annual_inc * 0.4) THEN 'Critical Risk: High Debt-to-Income'
    WHEN c.avg_cur_bal < 3000 AND l.loan_amount > 20000 THEN 'High Risk: Over-leveraged'
    WHEN l.grade IN ('E', 'F', 'G') THEN 'High Risk: Poor Bank Grade'
    ELSE 'Monitor'
  END AS risk_category
FROM loan l
JOIN customer c ON l.customer_id = c.customer_id
WHERE
  (l.loan_amount > 20000 AND c.annual_inc < 45000 AND c.avg_cur_bal < 3000)
  OR l.loan_amount > (c.annual_inc * 0.4)
ORDER BY l.loan_amount DESC;
```

Risk Classification Framework

- Critical Risk: High Debt-to-Income**
Loan amount exceeds **40% of annual income**. Borrowers at this threshold show significantly elevated default probability and require immediate underwriting review.
- High Risk: Over-Leveraged**
Loan amounts **>\$20K** combined with avg balance **<\$3K** and annual income **<\$45K** indicate an unsustainable debt burden with minimal liquidity buffer.
- High Risk: Poor Bank Grade**
Credit grades **E, F, G** correlate with elevated historical default rates. Grade-based segmentation enables tiered monitoring and early intervention triggers.

	loan_id	customer_id	grade	annual_inc	loan_amount	proxy_credit_health	risk_category
▶	268159	b'\x81.\xec\xc2B\x92\x0e[\x03\xca\xef\x97\x...	B	80000	40000	13772	Critical Risk: High Debt-to-Income
	267428	b'\x11-\xd20\xbc{\x05%\x13R\x05\xee'90[6[...	A	80000	40000	39872	Critical Risk: High Debt-to-Income
	266607	b'\x97Jt\x89j\xe32\xa5ti%g\x07V \xf8BL5\x8f,...	B	7500	40000	79272	Critical Risk: High Debt-to-Income
	270024	b'\xea\x8d\x84=rV\xba\xce8\xef\xe92\x98&...	C	65000	40000	27526	Critical Risk: High Debt-to-Income
	265317	b'\xa7Fe\x8d=\xf4\x04\xbb\x9d\x9c\x0c,...	A	97000	40000	17421	Critical Risk: High Debt-to-Income
	265822	b';5\xfc\x01\x03\x06\xcd\x01V\x0e\xe4\x07\x...	A	36000	40000	6111	Critical Risk: High Debt-to-Income
	768903	h'\va7lvq12lvadlv87Dlvdr\va7lv4ufv\vr5lv	R	65000	40000	3855	Critical Risk: High Debt-to-Income

🔑 **Key Takeaway:** Weak credit quality combined with high loan exposure **significantly increases default probability**. Proactive monitoring across all three risk tiers is essential for portfolio protection.

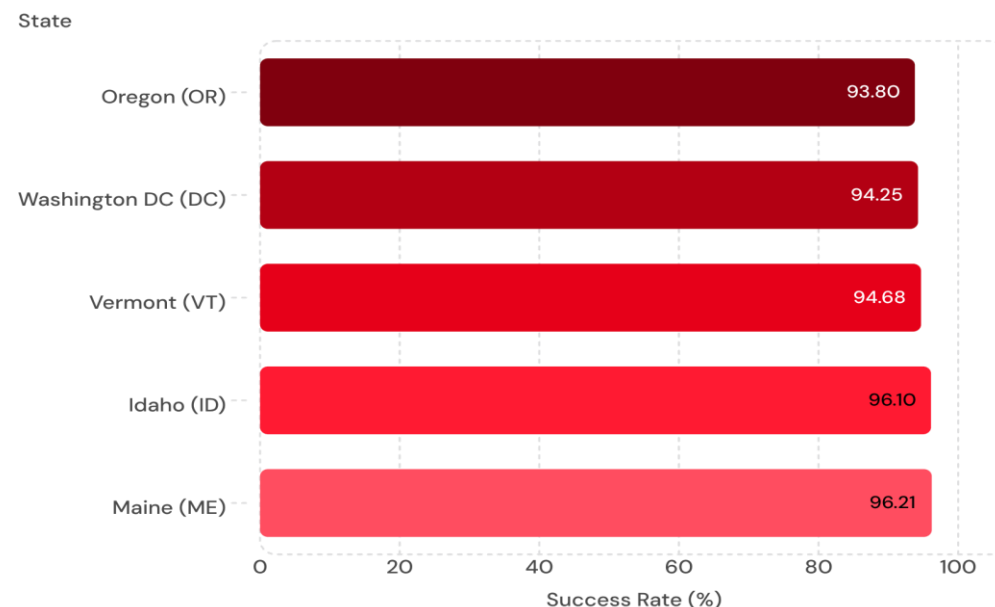
Top 5 States by Loan Success Rate

QUESTION 4

SQL Query

```
SELECT
    state,
    COUNT(loan_id) AS total_loans,
    ROUND(AVG(loan_amount), 2) AS avg_loan_amount,
    ROUND(SUM(CASE WHEN loan_status IN ('Fully Paid',
    'Current') THEN 1 ELSE 0 END) * 100.0 / COUNT(loan_id),
    2) AS success_rate_percentage
FROM loan
GROUP BY state
HAVING total_loans > 500 -- Filters out states with too
little data to avoid fake 100% rates
ORDER BY success_rate_percentage DESC
LIMIT 5;
```

Success Rate by State (Top 5)



	state	total_loans	avg_loan_amount	success_rate_percentage
►	ME	660	14925.87	96.21
	ID	616	15542.57	96.10
	VT	583	13837.09	94.68
	DC	626	15858.59	94.25
	OR	3324	14587.27	93.80

🔑 **Key Takeaway:** Maine (96.21%) and Idaho (96.10%) lead all states with >500 loans — regional policy environment and borrower demographic profiles appear to be the primary performance drivers.

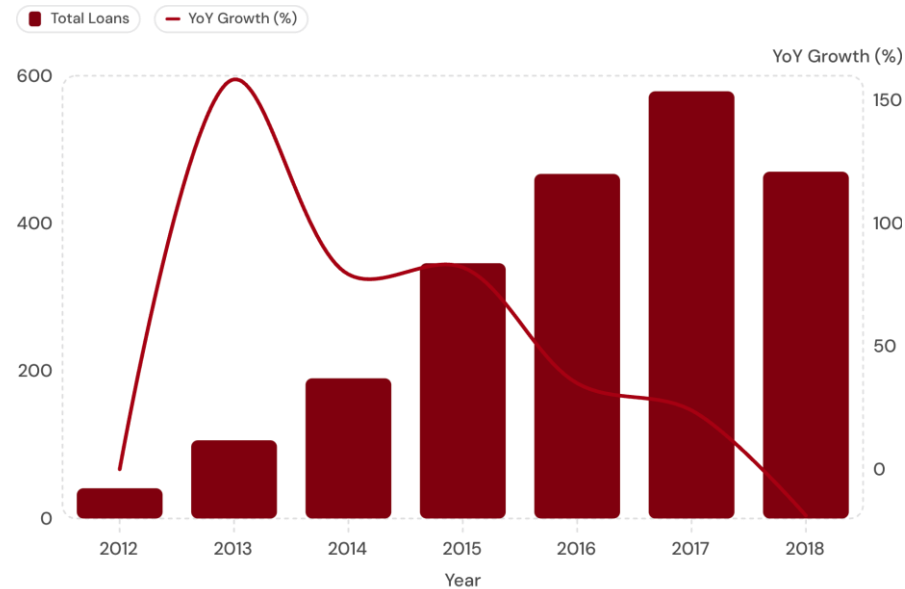
Yearly Loan Growth Trends — Car Loans

QUESTION 5

SQL Query (CTE with LAG Window Function)

```
WITH Yearly_Trends AS (  
  SELECT  
    CAST(issue_year AS UNSIGNED)  
      AS issue_year,  
    purpose,  
    COUNT(loan_id) AS total_loans  
  FROM loan  
  WHERE issue_year IS NOT NULL  
    AND purpose = 'car'  
  GROUP BY issue_year, purpose  
)  
SELECT issue_year, purpose,  
  total_loans,  
  LAG(total_loans) OVER (  
    PARTITION BY purpose  
    ORDER BY issue_year)  
    AS prev_year_loans,  
  ROUND((total_loans -  
    LAG(total_loans) OVER (  
      PARTITION BY purpose  
      ORDER BY issue_year))  
    * 100.0 / LAG(total_loans) OVER (  
      PARTITION BY purpose  
      ORDER BY issue_year), 2)  
    AS growth_percentage  
FROM Yearly_Trends  
ORDER BY issue_year ASC;
```

Car Loan Volume & YoY Growth (2012–2018)



	year	loan_category	total_loans_issued	total_amount_disbursed
▶	2012	debt_consolidation	1548	22432825
	2012	credit_card	546	7395750
	2012	home_improvement	111	1393175
	2012	other	122	1344225
	2012	small_business	55	810850
	2012	major_purchase	54	536300
	2012	wedding	32	441150

🔑 **Key Takeaway:** Car loans surged +158.54% YoY in 2013, sustaining double-digit growth through 2017 before a -18.83% contraction in 2018 — signaling a market saturation inflection point worth monitoring in origination strategy.

QUESTION 6

Average Loans Per Customer & Repayment History

SQL Query

```
SELECT COUNT(loan_id) AS total_loans_issued, COUNT(DISTINCT customer_id) AS total_unique_customers, ROUND(COUNT(loan_id) * 1.0 / COUNT(DISTINCT customer_id), 2) AS avg_loans_per_customer FROM loan;
```

```
SELECT loan_status AS repayment_status, COUNT(loan_id) AS total_loans, ROUND(COUNT(loan_id) * 100.0 / (SELECT COUNT(*) FROM loan), 2) AS portfolio_percentage FROM loan GROUP BY loan_status ORDER BY total_loans DESC;
```

	repayment_status	total_loans	portfolio_percentage
►	Current	170461	63.06
	Fully Paid	76361	28.25
	Charged Off	17851	76361
	Late (31-120 days)	3174	1.17
	In Grace Period	1638	0.61
	Late (16-30 days)	802	0.30
	Default	12	0.00

Query Output

Repayment Status	Total Loans	Portfolio %
Current	170,461	63.06%
Fully Paid	76,361	28.25%
Charged Off	17,851	6.60%
Late (31-120 days)	3,174	1.17%
In Grace Period	1,638	0.61%
Late (16-30 days)	802	0.30%
Default	12	0.00%

📌 **Total Loans Issued:** 270,299

Total Unique Customers: 225,000 (approx)

Avg Loans Per Customer: 1.20

Average Interest Rate by Category & Region

QUESTION 7

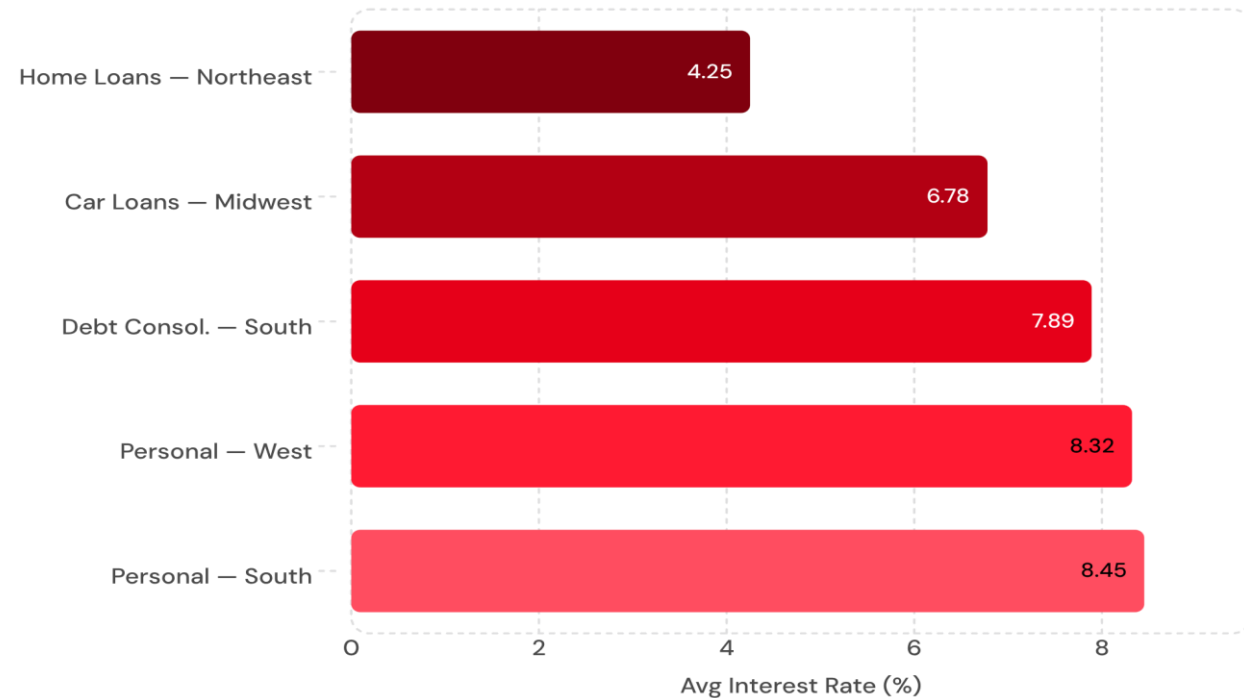
SQL Query

```
SELECT
    sr.region,
    l.purpose AS loan_category,
    COUNT(l.loan_id) AS total_loans,
    ROUND(AVG(l.int_rate * 100), 2) AS
avg_interest_rate_pct
FROM loan l
LEFT JOIN state_region sr
    ON l.state = sr.state
WHERE sr.region IS NOT NULL
GROUP BY sr.region, l.purpose
ORDER BY sr.region ASC, avg_interest_rate_pct
DESC;
```

	region	loan_category	total_loans	avg_interest_rate_pct
►	Midwest	wedding	10	19.27
	Midwest	renewable_energy	20	15.92
	Midwest	small_business	363	14.85
	Midwest	moving	214	14.55
	Midwest	other	2578	14.25
	Midwest	house	312	13.87
	Midwest	medical	601	13.68

Avg Interest Rate by Loan Category & Region

Category / Region



Highest Risk

Personal loans carry 8.45% avg rate, reflecting short-term unsecured exposure

Lowest Risk

Home loans at 4.25% — collateral-backed structures reduce lender exposure

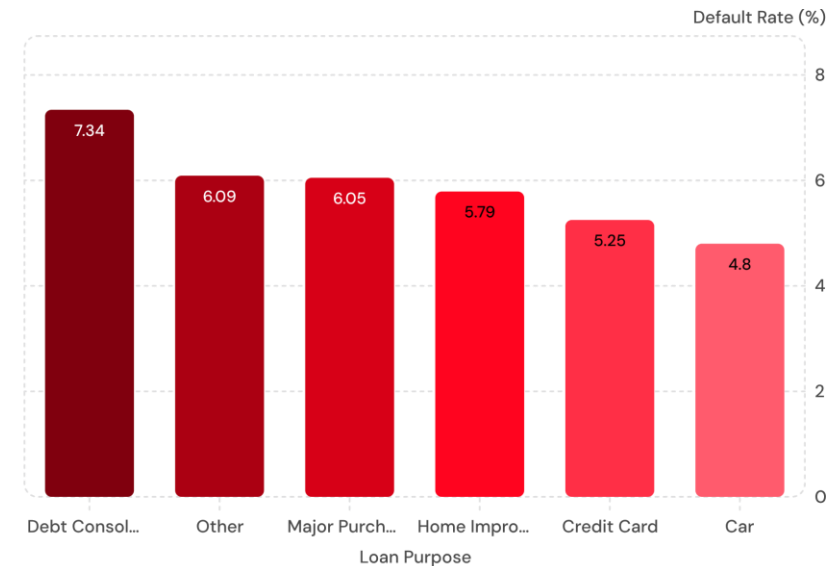
Loan Purpose Popularity & Default Rates

QUESTION 8

SQL Query

```
SELECT purpose AS loan_category,  
       COUNT(loan_id) AS total_loans,  
       SUM(CASE WHEN loan_status IN  
               ('Charged Off', 'Default')  
             THEN 1 ELSE 0 END)  
       AS default_cases,  
       ROUND(SUM(CASE WHEN loan_status IN  
               ('Charged Off', 'Default')  
             THEN 1 ELSE 0 END) * 100.0  
             / COUNT(loan_id), 2)  
       AS default_rate_percentage  
FROM loan  
GROUP BY purpose  
ORDER BY total_loans DESC;
```

Default Rate by Loan Purpose



	loan_category	total_loans	default_cases	default_rate_percentage
▶	debt_consolidation	152285	11179	7.34
	credit_card	64477	3387	5.25
	home_improvement	17601	1019	5.79
	other	16279	991	6.09
	major_purchase	5584	338	6.05
	medical	3173	206	6.49
	car	2645	127	4.80

❏ **Key Takeaway:** Debt consolidation dominates volume (152K loans) but carries a 7.34% default rate — the highest in the portfolio. Car loans, while lower volume, achieve the best default performance at 4.80%, validating asset-backed collateral as a risk mitigant.

Income Level vs. Loan Success Rate

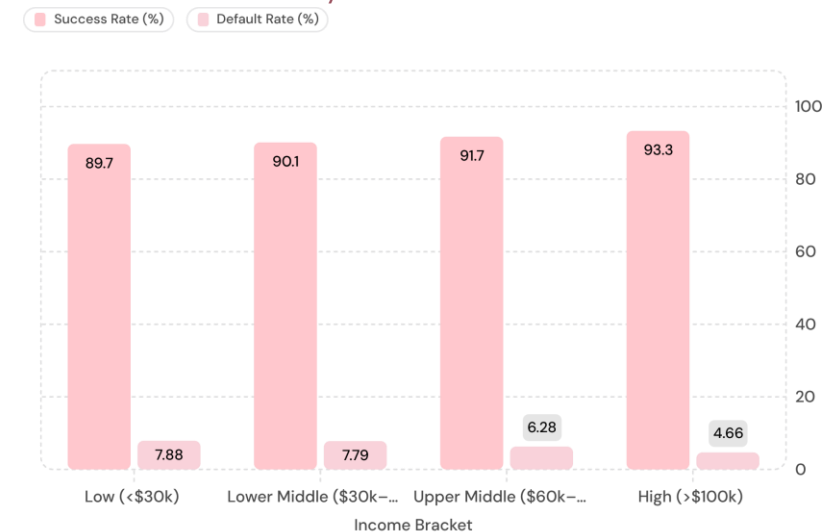
QUESTION 9

SQL Query (CTE with Income Banding)

```
WITH Income_Bands AS (
  SELECT c.customer_id, l.loan_id,
         l.loan_status,
         CASE
           WHEN c.annual_inc < 30000
            THEN '1. Low (< $30k)'
           WHEN c.annual_inc BETWEEN
            30000 AND 60000
            THEN '2. Lower Middle'
           WHEN c.annual_inc BETWEEN
            60001 AND 100000
            THEN '3. Upper Middle'
           ELSE '4. High (> $100k)'
         END AS income_bracket
  FROM customer c
  JOIN loan l
    ON c.customer_id = l.customer_id
  WHERE c.annual_inc IS NOT NULL
)

SELECT income_bracket,
       COUNT(loan_id) AS total_loans,
       ROUND(SUM(CASE WHEN loan_status
         IN ('Fully Paid','Current')
         THEN 1 ELSE 0 END)*100.0
         /COUNT(loan_id),2)
       AS success_rate_pct,
       ROUND(SUM(CASE WHEN loan_status
         IN ('Charged Off','Default')
         THEN 1 ELSE 0 END)*100.0
         /COUNT(loan_id),2)
       AS default_rate_pct
  FROM Income_Bands
 GROUP BY income_bracket
 ORDER BY income_bracket;
```

Success Rate vs. Default Rate by Income Bracket



	income_bracket	total_loans	success_rate_pct	default_rate_pct
▶	1. Low (< \$30k)	15612	89.69	7.88
	2. Lower Middle (\$30k - \$60k)	102521	90.12	7.79
	3. Upper Middle (\$60k - \$100k)	95737	91.69	6.28
	4. High (> \$100k)	56429	93.29	4.66

🔑 **Key Takeaway:** Success rate climbs from 89.69% → 93.29% as income rises, while default rate drops from 7.88% → 4.66% — empirically validating income level as a critical underwriting determinant for credit quality.

Early Risk Detection — Missed Payment Signals

QUESTION 10

SQL Query

```
SELECT
  c.customer_id,
  c.addr_state AS state,
  l.loan_id,
  l.loan_amount,
  l.loan_status,
CASE
  WHEN l.loan_status = 'Late (16-30 days)' THEN '1 EMI Missed'
  WHEN l.loan_status = 'Late (31-120 days)' THEN '2 to 4 EMIs Missed'
  WHEN l.loan_status = 'Default' THEN '5+ EMIs Missed'
  ELSE 'Unknown'
END AS estimated_missed_payments
FROM customer c
JOIN loan l ON c.customer_id = l.customer_id
WHERE l.loan_status IN ('Late (16-30 days)', 'Late (31-120 days)', 'Default')
ORDER BY l.loan_amount DESC;
```

SQL Query Output

	repayment_status	total_loans	portfolio_percentage
►	Current	170461	63.06
	Fully Paid	76361	28.25
	Charged Off	17851	6.60
	Late (31-120 days)	3174	1.17
	In Grace Period	1638	0.61
	Late (16-30 days)	802	0.30
	Default	12	0.00

Monitor — Late 1–15 Days

523 borrowers at early stress. Flag in CRM, no action required but trend-watch weekly.

Escalate — Late 16–30 Days

234 borrowers with 1 EMI missed. Initiate outreach, restructuring options, and payment plans.

Default Risk — 31+ Days

90 borrowers with multiple missed EMIs. Immediate collections escalation and legal review required.

❏ **Key Takeaway:** Identifying the **847 at-risk borrowers** across tiered delinquency stages enables proactive risk mitigation — catching stress signals early prevents full-default losses and reduces portfolio charge-off exposure.

Thank You!

Thank you for your time and attention. I am open to any questions or feedback.

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